

Problem 18.11

The present value of a perpetuity-immediate where the payment is P is 1,000 less than the present value of a perpetuity-due where the payment is P . Calculate P .

Solution

The present value of a perpetuity-immediate with payment P is

$$Pa_{\infty|}.$$

The present value of a perpetuity-due with payment P is

$$P\ddot{a}_{\infty|}.$$

We are told that the perpetuity-immediate value is 1,000 less than the perpetuity-due value, so

$$Pa_{\infty|} = P\ddot{a}_{\infty|} - 1000.$$

Equivalently,

$$1000 = P(\ddot{a}_{\infty|} - a_{\infty|}).$$

For a perpetuity,

$$a_{\infty|} = v + v^2 + v^3 + \dots$$

and

$$\ddot{a}_{\infty|} = 1 + v + v^2 + v^3 + \dots.$$

Thus,

$$\ddot{a}_{\infty|} - a_{\infty|} = 1.$$

Therefore,

$$1000 = P(1).$$

Hence,

$$\boxed{P = \$1,000}.$$